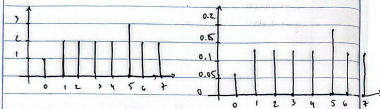


Al. 1)
$$\begin{bmatrix} 0 & 1 & 2 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \\ 5 & 6 & 7 \end{bmatrix}$$

Intensity 0 1 2 3 4 5 6 7
Histogram 1 2 2 2 2 3 2 2
N. Histogram 0.0625 0.125 0.125 0.125 0.125 0.1875 0.125 0.125



2)
$$S = T(r)$$

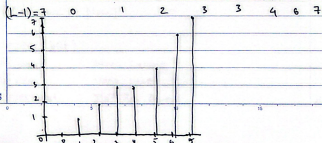
$$S = \left[\frac{L-1}{\max} \sum_{j=0}^r \frac{h(j)}{N} \right]$$

$$h(j) = \frac{N(j)}{N}$$

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0 1 2 3 4 5 6 7
cat 1 3 5 7 9 11 14 16
0.0625 0.1875 0.3125 0.4375 0.5625 0.6875 0.8125 0.9375



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3) Median is average of 8th and 9th value (16 pixels)

$$K = \frac{4+4}{2} = 4$$

→ threshold = 4

0 1 2 3 4 5 6 7
0 0 0 0 0 1 1 1

→ Binary image:

$$I = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

4) Negative images

$$s = \text{Intensity max} - r$$

$$I = \begin{bmatrix} 7 & 6 & 6 & 5 \\ 5 & 4 & 4 & 3 \\ 3 & 2 & 2 & 1 \\ 2 & 1 & 0 & 0 \end{bmatrix}$$

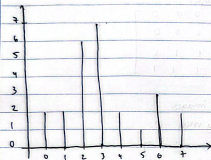
HAPLUS

A2.

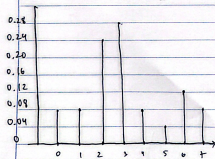
1)

Intensity:	0	1	2	3	4	5	6	7
Hist	2	2	6	7	2	1	3	2
N.Hist	0.08	0.08	0.24	0.28	0.08	0.04	0.12	0.08

Histogram



N. histogram.



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- 2) Mode of gray-level distribution = 3 (appear 7 times).
 3) $k = 3$

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

- 4) Foreground $B(x,y) = 1$: 8 pixels
 Background $B(x,y) = 0$: 17 pixels.
 5) Not very well cause the background class pixels (low intensity) are ~~double~~ double twice as much the foreground (high intensity) ones.

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